Types & Decisions

Classwork: let vs const, Types, Operators & if

Name: 　　　　　　　　**Date:**

Last week your code stored things. Today it makes decisions! Work through each mission in order. Type the code yourself — don't copy-paste — and always predict the output BEFORE you press Run.

Before You Start — Setup (about 3 min)

Everything today happens in your browser. No installing anything!

1. Go to typescriptlang.org/play
2. Delete any sample code so you start with an empty file.
3. Your code goes on the LEFT. Press Run to see output on the RIGHT.

TIP
See a red squiggly underline? That's TypeScript telling you something is wrong BEFORE you even run it. Hover your mouse over it to read the message.

Mission A — Warm-Up (about 8 min)

Let's rebuild what you already know, then check your memory.

Type this code:

```
const food = "momo"
const num = 10
console.log(food)
console.log(num)
```

What printed in the output panel?

Now PREDICT before you run — write your guess first, then try it:

Predict, then run:

```
console.log(10 + 1)
console.log("10" + 1)
```

My prediction for line 1:

My prediction for line 2:

What actually happened? Were you right?

Mission B — let vs const (about 10 min)

const is LOCKED. let can CHANGE. Let's prove it.

1. Type this and run it:

Type this code:

```
let score = 0
score = 100
console.log(score)
```

What number printed?

2. Now break it on purpose — change let to const and run again.

What did the red error message say?

In your own words: when should you use let instead of const?

Mission C — Add Types (about 12 min)

Time to make it real TypeScript! Add a type after the variable name, using a colon.

Type this code:

```
const myName: string = "put your name here"
let myAge: number = 12
let likesPizza: boolean = true
console.log(myName, myAge, likesPizza)
```

Did it run with no errors? (yes / no)

1. Now BREAK each one on purpose and write down the error you get:

Break it #1:

```
let myAge: number = "twelve"
```

What error appeared?

Break it #2:

```
let likesPizza: boolean = 42
```

What error appeared?

CONCEPT
TypeScript caught your mistake BEFORE running the program. That's the whole point of types — the computer checks your work for you!

Mission D — Template Literals (about 8 min)

Backticks ` ` let you drop variables straight into text with ${ }. Much nicer than gluing with +.

Type this code:

```
const myName: string = "your name"
const myAge: number = 12
console.log(`Hi! I am ${myName} and I am ${myAge}.`)
```

What did it print?

Now build a PROFILE CARD. Make 4 variables about yourself (with types!) and print them in one nice sentence using a template literal.

Write your code here:

What did your profile card print?

Mission E — Math & Comparisons (about 10 min)

Operators do math. Comparisons ask true/false questions.

Type this code:

```
console.log(10 + 3)
console.log(10 - 3)
console.log(10 * 3)
console.log(10 / 2)
console.log(10 % 3)
```

What did 10 % 3 print? What do you think % does?

Now comparisons. PREDICT each answer (true or false) before running:

Predict, then run:

```
console.log(5 === 5)
console.log(10 > 20)
console.log(7 !== 3)
console.log(4 < 4)
```

My predictions (in order):

How many did you get right?

CONCEPT
Every comparison gives back a boolean — true or false. That's exactly what an if statement needs!

Mission F — if / else (about 12 min)

Now your program can DECIDE what to do.

Type this code:

```
let score: number = 120

if (score > 100) {
  console.log("You win!")
} else {
  console.log("Try again!")
}
```

What printed?

1. Change score to 50 and run again. What printed this time?

Your answer:

Now write your OWN if/else. Idea: check if a number is even using % 2 === 0

Write your code here:

Did it work? What did it print?

Mission G — Quick Quests (about 10 min)

Small challenges! Check the box as you finish each one.

- [ ] Make a variable with a type for your favorite food, then print it.
- [ ] Use let to make a score, add 50 to it, and print the new score.
- [ ] Print a sentence about yourself using a template literal.
- [ ] Write an if statement that prints something when a number is bigger than 10.
- [ ] Write an if/else that says whether a number is even or odd.
- [ ] Use % to find the remainder when 25 is divided by 4.
- [ ] Make a boolean variable and use it directly inside an if statement.
- [ ] STRETCH: add an else if so your program has THREE possible answers.

Which quest was your favorite, and why?

Mission H — Mini Project: Your Own Program (about 10 min)

Put it ALL together. Your program must use: a typed variable, a let that changes, an operator, and an if/else. Ideas: a game score checker, a grade calculator, an 'am I old enough?' checker.

What does your program do?

Write your code here:

What does it print?

Wrap-Up — Reflect (about 4 min)

What was the trickiest part of today?

What would you want your program to do next week?